

Energy Forms and Transfers

Energy and its
Forms 1

I can...

- Illustrate the types of energy and energy transformation.
- illustrate how energy can be transformed from one type to another.

What is energy?

Energy – measurement of the ability to make mass move

➤ **Energy** is the ability to do **work**

➤ In physics, **work** = **Force** · **displacement**

➤ **Force (F)** – push or pull, measured in **Newtons**

➤ **displacement (d)** – how far something moves, measured in **meters**

➤ SI unit: **Joule (J)** = 1 **N**·**m** (1 **Newton meter**)

Quantity	Definition	Units	Unit Name	Abbreviation
energy	= Fd	$N \cdot m$	joule	J

“Using” Energy



Energy never goes away; it only changes forms.

Law of Conservation of Energy – Energy cannot be created nor destroyed; only converted from one form to another.

- All the energy in the universe was present at the Big Bang.
- The universe is a **closed system**, meaning nothing can leave it.
- A cup of coffee is an **open system**, the heat goes into the air, but it never leaves the universe.

What Kinds of Energy are there?



mechanical



electrical



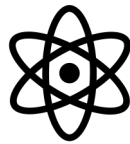
magnetic



gravitational



chemical



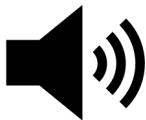
nuclear



elastic



wave



sound



radiant



rest



thermal

Energy can be stored in many forms.

Energy store	Description	Examples
Magnetic 	The energy stored when repelling poles have been pushed closer together or when attracting poles have been pulled further apart.	Fridge magnets, compasses, maglev trains which use magnetic levitation.
Internal (thermal) 	The total kinetic and potential energy of the particles in an object, the vibrations of particles. In hotter objects, the particles have more internal energy and vibrate faster.	Human bodies, hot coffees, stoves or hobs. Ice particles vibrate slower, but still have energy.
Chemical 	The energy stored in chemical bonds, such as those between molecules.	Foods, muscles, electrical cells.
Kinetic 	The energy of a moving object.	Runners, buses, comets.
Electrostatic 	The energy stored when repelling charges have been moved closer together or when attracting charges have been pulled further apart.	Thunderclouds, Van De Graaff generators.
Elastic potential 	The energy stored when an object is stretched or squashed.	Drawn catapults, compressed springs, inflated balloons.
Gravitational potential	The energy of an object at height.	Aeroplanes, kites, mugs on a table.
Nuclear 	The energy stored in the nucleus of an atom.	Uranium nuclear power, nuclear reactors.

Energy Store	Definition
chemical	The energy stored in a moving object.
elastic potential	The energy stored when an object is lifted in a gravitational field.
kinetic	The energy stored when an object has been stretched or squashed.
thermal	The energy stored when repelling poles have been pushed closer together or when attracting poles have been pulled further apart.
magnetic	The energy stored in the bonds of a substance or group of substances.
electrostatic	The energy stored when repelling charges have been pushed closer together or when attracting charges have been pulled further apart.
gravitational potential	The total energy of the particles that make up an object. The more energy there is in this store, the higher the temperature of the object.
nuclear	The energy stored in the nucleus of an atom.

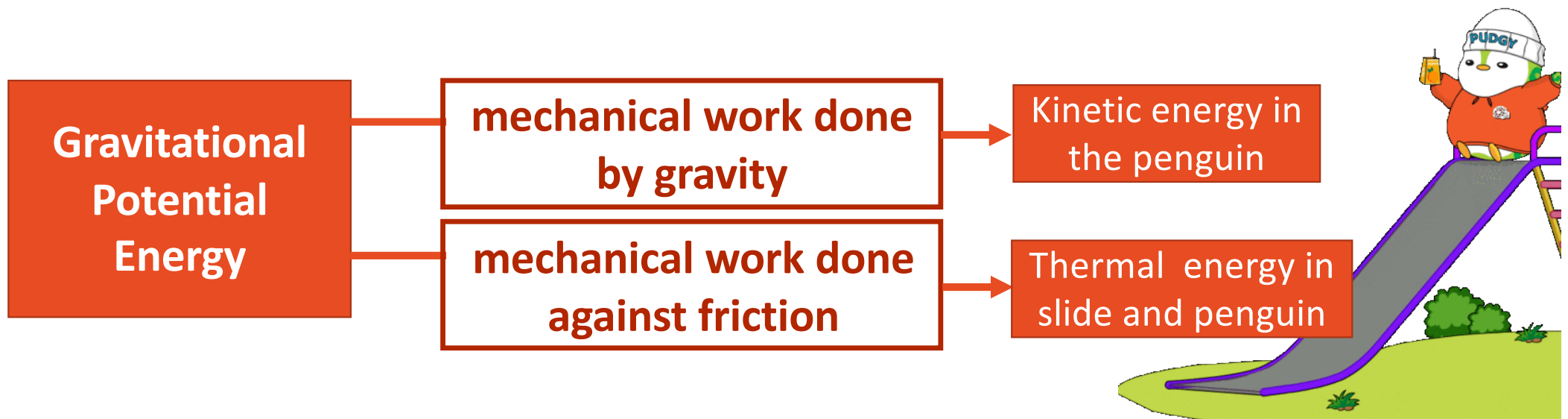
Energy Transfer

Energy can be converted from one form to another.

- mechanical work - a force moving an object through a distance
- electrical work – charged particles (electrons) moving
- heating - due to difference in temperature, or the average vibration of particles
- radiation - energy transferred as an electromagnetic wave



You Try



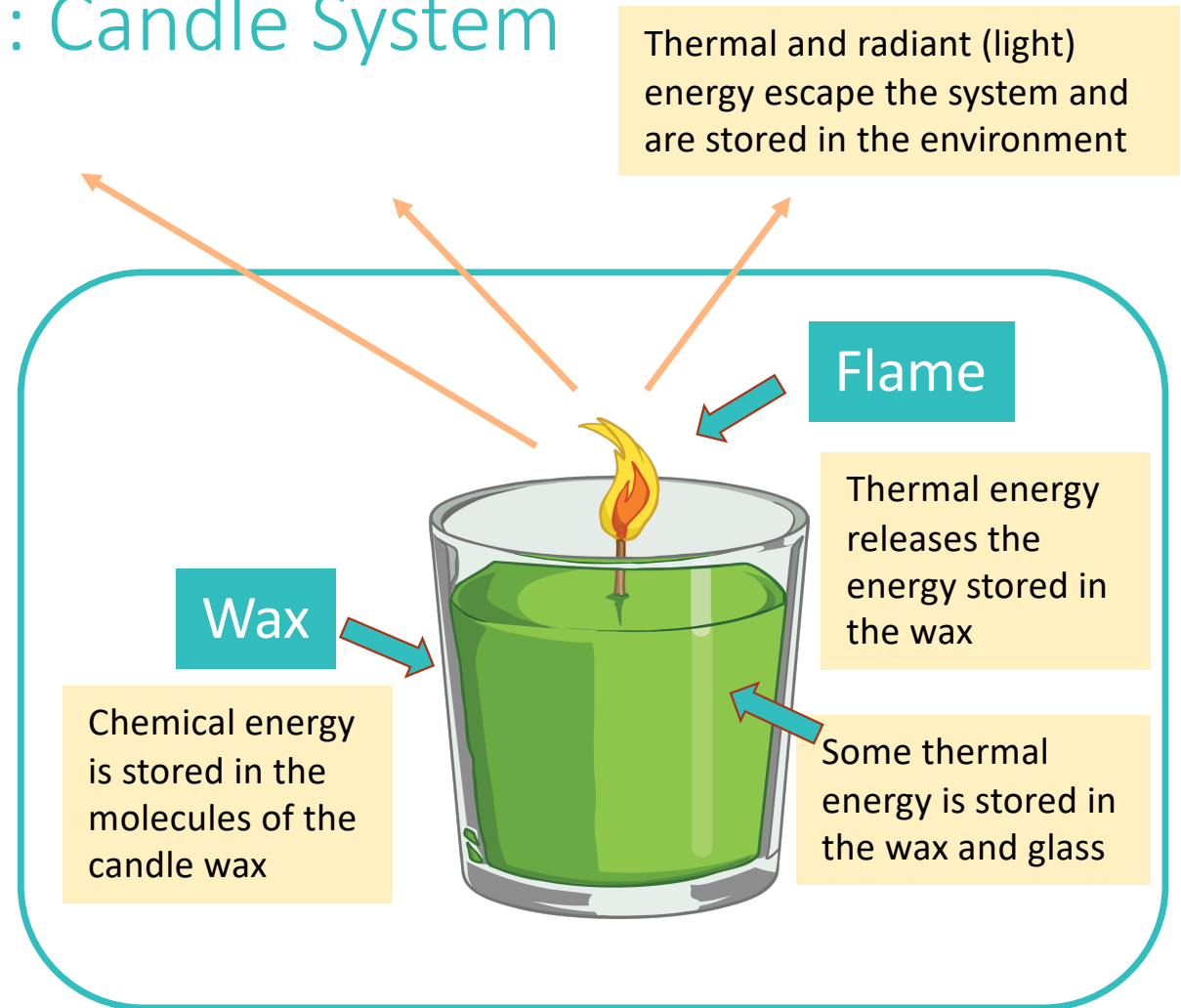
Example: Candle System

1. Define the system

2. Find the energy form at the beginning

3. Find how the energy is being transferred

4. Find the energy forms at the end



You try: Battery-Powered Fan

1. Define the system

2. Find the energy form at the beginning

3. Find how the energy is being transferred

4. Find the energy forms at the end

